# Chapter 4: Medication management and drug stewardship in CKD

Medication management is an important component of the care of people with CKD. Medications can be highly beneficial, but some may be toxic, are excreted by the kidney, may have narrow therapeutic windows, or may have no proven clear evidence of benefit or indication in people with CKD.

Drug stewardship refers to the effective, safe, and sustainable use of medications by all staff and physicians, encompassing the whole cycle of medication use. Medications need to be prescribed responsibly, monitored for efficacy and safety, and when they do not or no longer serve their intended purpose, discontinued. This chapter discusses key concepts in the processes of drug stewardship in people with CKD. It is beyond the scope of this guideline to list all the medications that may have altered risks/benefits in people with CKD. Such information is widely available in documents that may exist at local, regional, or national bodies (e.g., British National Formulary: www.bnf.org) and in textbooks of pharmacology. However, we describe case examples to highlight the key classes of commonly prescribed medications in people with CKD. This guidance is based on knowledge of pharmacology that has universal relevance. In many cases, knowledge of altered risks/benefits of medications comes, however, from observational studies and case reports from routine care.

## 4.1 Medication choices and monitoring for safety

Abnormal kidney function results in alteration in pharmacokinetics and pharmacodynamics, and for people with CKD, as the GFR worsens, so does the prevalence of polypharmacy and comorbidities.[725] People with CKD are at increased risk of medication errors and inappropriate prescribing (noted to be up to 37% in ambulatory outpatient studies and up to 43% in long-term care studies[726,727]). Thus, improved understanding and collaboration with pharmacists in developing care plans and medication review is strongly recommended.

People with CKD have reduced ability to excrete medications and/or their metabolites (which may increase adverse event risk or exaggerate/diminish efficacy) and increased sensitivity to medications (e.g., those bound to albumin in hypoalbuminemic states such as nephrotic syndrome). Additional issues include nephrotoxicity, diminished tolerance of side effects in the context of coexisting comorbidities or older age, and lack of adequate evidence for either benefit or harm of specific compounds, due to historical exclusion of people with (advanced) CKD from most clinical trials.[727,728]

As in all medical decision-making, healthcare providers should consider the indication, benefit-risk profile, and potential nephrotoxicity while balancing accessibility, availability, local health policies, cultural practices, affordability, and patient preferences. Where available, consultation with pharmacists as part of the multidisciplinary team is encouraged to assure optimized comprehensive medication management and to improve pharmacoequity.

**Practice Point 4.1.1: People with CKD may be more susceptible to the nephrotoxic effects of medications. When prescribing such medications to people with CKD, always consider the benefits versus potential harms.**

Between 18%–20% of people with CKD G3–G5 receive at least one potentially inappropriate nephrotoxic medication annually, primarily NSAIDs, nephrotoxic antivirals, and bisphosphonates.[729] Nephrotoxic medications may be indicated in people with CKD if expected benefits exceed potential harms.[730] However, whenever possible, healthcare providers should strive to use non-nephrotoxic alternatives. Common nephrotoxic medications to be aware of and potential alternatives that could be prescribed instead are listed in Table 31.[725,731–738] Although some nephrotoxic medications have viable alternatives, the alternatives may be less potent or there is limited comparison data on clinical outcomes, safety, and cost-effectiveness.

**Practice Point 4.1.2: Monitor eGFR, electrolytes, and therapeutic medication levels, when indicated, in people with CKD receiving medications with narrow therapeutic windows, potential adverse effects, or nephrotoxicity, both in outpatient practice and in hospital settings.**

Ensuring a safe use of medication requires careful monitoring for adverse effects and efficacy. A key example includes the need to monitor potassium and creatinine during the initial weeks of treatment with ACEi and ARBs (Figure 21).[23] Medications such as gentamicin and vancomycin have a narrow therapeutic range, with higher trough levels commonly associated with AKI, and so require close monitoring of GFR and medication levels during prolonged treatment.[731] Other medications, such as lithium or methotrexate, require at least annual monitoring of creatinine to evaluate potential risks of nephrotoxicity.

**Practice Point 4.1.3: Review and limit the use of over-the-counter medicines and dietary or herbal remedies that may be harmful for people with CKD.**

**Table 31 | Key examples of common medications with documented nephrotoxicity and, where available, selected non-nephrotoxic alternatives**

| Nephrotoxic medication | Potential non-nephrotoxic alternatives |
|---|---|
| **Analgesics** | |
| NSAIDs: nephrotoxic effects include a decrease in GFR through a reduction in prostaglandin-dependent kidney blood flow, allergic interstitial nephritis (AIN), and nephrotic syndrome[725] | Acetaminophen |
| **Antimicrobials** | |
| Aminoglycosides: accumulates in the proximal tubular cells and disrupts phospholipid metabolism, resulting in cell apoptosis and acute tubular necrosis (ATN)[731,732] | Cephalosporins and carbapenems |
| Vancomycin: unclear cause of nephrotoxicity, but likely related to ATN and possible AIN[731,732] | Linezolid and daptomycin[731] |
| Sulfamethoxazole-trimethoprim: AIN, ATN, crystalluria within the distal convoluted tubule and reversible inhibition of tubular creatinine secretion[731] | Clindamycin + primaquine, pentamidine, and atovaquone |
| **Gastrointestinal medications** | |
| Proton pump inhibitors: may result in AKI and CKD due to tubulointerstitial nephritis and AIN[733,734] | H2-receptor antagonists |
| **Cardiovascular medications** | |
| Warfarin: glomerular hemorrhage, oxidative stress causing kidney tubular damage, and direct effects on kidney vascular calcification by vitamin K–dependent alterations of matrix Gla protein[735,736] | Non–vitamin K antagonist oral anticoagulants |
| **Other** | |
| Lithium: nephrogenic diabetes insipidus as well as CKD from chronic tubulointerstitial nephropathy[737] | Aripiprazole, lamotrigine, quetiapine, valproate |

CKD, chronic kidney disease; GFR, glomerular filtration rate; NSAID, nonsteroidal anti-inflammatory drug.

Kidney disease can be induced or accelerated by the use of certain over-the-counter (OTC) medications, herbal remedies, and other dietary supplements. One of the most used class of OTC analgesic medications is NSAIDs. NSAIDs are associated with interstitial nephritis, analgesic nephropathy, and hypertension.[739] Indiscriminate chronic OTC NSAID use has been associated with a higher risks of kidney failure compared with nonuse and should be discouraged.[740–743] However, judicious NSAID use, under careful supervision of a nephrologist, may be preferred to other pain medications such as opioids that have stronger associations with adverse events.[744,745] PPIs are also common OTC medications in some countries that have been associated with AKI and CKD due to tubulointerstitial nephritis and acute interstitial nephritis.[733,734]

The use of herbal compounds remains highly prevalent in some countries and cultures.[746] These products are often used in an unmonitored setting without the input of healthcare providers. Many of these remedies are composed of natural compounds with complex active ingredients that have not been evaluated in people with CKD and/or that may lead to many different adverse effects. The frequency of CKD associated with herbal remedy use is not known and is likely different in different parts of the world, depending on local availability and reasons for use. Examples include aristolochic acid nephropathy or nephrotoxicity due to alkaloid compounds often found in Chinese herbal remedies.[747] However, cases of nephrotoxicity have been reported for many other herbal remedies globally.[746,748,749] The potential toxicity of herbal remedies may be enhanced by coexisting volume depletion and by other illness or medication use.

Dietary supplements are readily available in most countries around the world and are usually not classified as OTC medications. Because of this, their regulation for identity and safety can vary widely. Although laws pertaining to dietary supplement labeling prohibit specific claims for the treatment or prevention of disease, these products are widely used as "alternative" or "complementary" therapy. Patients and providers often assume that these products are at least safe and possibly effective. Their pharmacokinetics may be unknown and potential toxicity unstudied. Classic examples include creatine supplements used for body building that have been associated with allergic interstitial nephritis.[750,751] Another example is vitamin C (ascorbic acid) supplements, which in excess can lead to tubular calcium oxalate crystal deposition.[752]

Healthcare providers are encouraged to routinely inquire about the use of herbal remedies and recommend stopping any unprescribed alternative remedy that may pose a threat for (kidney) health. Figure 45[747,753,754] lists common herbal remedies and dietary supplements arranged by the countries where the adverse effects were reported to increase awareness and facilitate discussions.

### Special considerations

***Global access to medications.*** Access to medications varies globally. Approximately 30% of the world population lacks timely access to quality medications. The International Society of Nephrology (ISN) reports that only 35% of patients in low-resource settings have access to ACEi/ARBs, statins, and

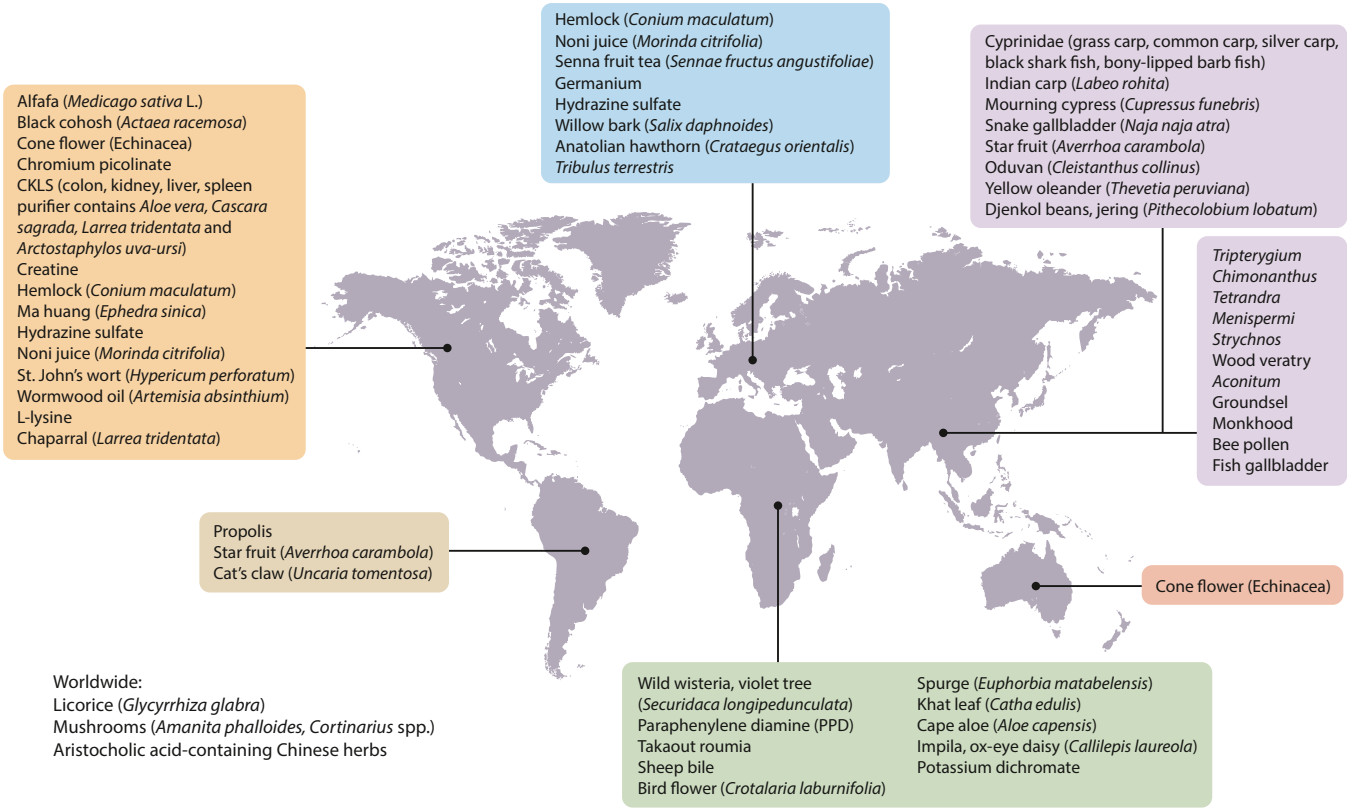


**Figure 45 | Selected herbal remedies and dietary supplements with evidence of potential nephrotoxicity, grouped by the continent from where the reports first came.** Data from Yang B, Xie Y, Guo M, et al. Nephrotoxicity and Chinese herbal medicine. *Clin J Am Soc Nephrol*. 2018;13:1605–1611[747]; Gabardi S, Munz K, Ulbricht C. A review of dietary supplement-induced renal dysfunction. *Clin J Am Soc Nephrol*. 2007;2:757–765[753]; Perazella MA. Pharmacology behind common drug nephrotoxicities. *Clin J Am Soc Nephrol*. 2018;13:1897–1908.[754]

insulin.[755] There are also numerous barriers to additional important medications for the management of CKD complications, such as erythropoietin analogs, iron infusion, and phosphate or potassium binders.

There are growing concerns regarding the use of falsified and substandard medications in low- to lower-middle–income countries as they pose potential harm, particularly to those people at risk of and with CKD. Patients and their families should be aware that medication falsification is often associated with illicit internet supply. Many vulnerable communities and people with low health literacy and those in countries with less rigorous regulatory systems are more at risk of medication falsification. Therefore, increased global awareness is important, and people with CKD should be provided with appropriate education and follow-up with relevant support in accordance with local health policies.

***Medications and pregnancy.***

**Practice Point 4.1.4: When prescribing medications to people with CKD who are of child-bearing potential, always review teratogenicity potential and provide regular reproductive and contraceptive counseling in accordance with the values and preferences of the person with CKD.**

When pregnancy is not desired, we note that while the effect of different forms of contraception on GFR is unknown,[756] oral contraceptives are associated with increased BP and hypertension.[757] Nonoral hormonal contraceptives have a less clear impact on BP.[757]

Pregnancy may pose a risk of CKD progression for people with established CKD. In addition, some recommended medications to slow or prevent CKD progression are teratogenic (such as ACEi/ARBs or mammalian target of rapamycin inhibitors) and discontinuation during pregnancy should be considered.[758] Some CKD-specific medications should be continued during pregnancies such as hydroxychloroquine, tacrolimus, cyclosporin, eculizumab, prednisone, azathioprine, colchicine, and intravenous immunoglobulin. A thorough medication chart review is necessary to replace teratogenic medications before conception, or whenever this is not possible, ensure a strict monitoring plan with cessation of potentially teratogenic medications at conception.[759] A similar approach should be undertaken during lactation recognizing that some medications suitable for use during pregnancy may not be appropriate for lactation, and vice versa.[760] Multidisciplinary care with obstetrics and potentially other subspecialty care is required before conception and throughout pregnancy and lactation.[54]

***Sex-specific aspects of medication use in CKD.*** Sex differences in medication safety and efficacy in people with CKD are understudied,[38,761,762] For example, sex differences in body weight and composition as well as physiological functions

may impact drug metabolism and response. Because drug dosages are often universal, women are more likely to consume higher doses in relation to their body weight,[136,763,764] and this could be associated with more adverse events.[136] In people with heart failure with reduced ejection fraction, observational studies show improved survival in women with lower doses of renin-angiotensin-aldosterone system (RAAS)-blocking medications, whereas men benefit from higher doses.[765,766] This may be related to lower RAAS activity in women compared with men.[767]

## 4.2 Dose adjustments by level of GFR

**Practice Point 4.2.1: Consider GFR when dosing medications cleared by the kidneys.**

Many medications and/or their active metabolites are excreted by the kidneys. Failure to properly account for the effect of GFR when designing appropriate drug-dosing regimens can predispose a person to treatment failure or adverse events.[725,728] Although guidelines for adjustment of the dosing regimen at varying severities of CKD provided by the manufacturer are widely available in pharmacopeias, textbooks, online references, or local procedures, there may be significant differences in information provided by these resources.[768]

**Practice Point 4.2.2: For most people and clinical settings, validated eGFR equations using SCr are appropriate for drug dosing.**

**Practice Point 4.2.3: Where more accuracy is required for drug-related decision-making (e.g., dosing due to narrow therapeutic or toxic range), drug toxicity, or clinical situations where eGFRcr estimates may be unreliable, use of equations that combine both creatinine and cystatin C, or measured GFR may be indicated.**

An assessment of GFR is important for guiding decisions related to the choice and dosing of medications. Section 1.2 addresses the accuracy of validated eGFR equations, as well as indications for the use of eGFRcr-cys or mGFR.

There is inconsistency between this guidance and those found in the package inserts or classic source references for drug dosing. Regulatory agencies have not universally required pharmacokinetics in abnormal kidney function for medication approval.[769] In addition, although the Cockcroft-Gault formula for estimating CrCl has been used in many past pharmacokinetic studies that serve as the basis for the drug dosing, there are multiple concerns with that equation. It was developed in an era when the need for standardization of creatinine measurements was not appreciated, women and individuals of Black race were not included, and there are concerns about use of weight, which can be impacted by edema or obesity.[770] However, to date, few studies have been conducted to compare different equations for eGFR in the context of drug dosing/kinetics, etc.

There is now a recognition by major regulatory agencies that "any contemporary, widely accepted, and clinically applicable estimating GFR equation is considered reasonable to assess GFR in pharmacokinetic studies."[770,771]

**Practice Point 4.2.4: In people with extremes of body weight, eGFR nonindexed for body surface area (BSA) may be indicated, especially for medications with a narrow therapeutic range or requiring a minimum concentration to be effective.**

For assessment of CKD, it is relevant to compare the GFR according to a standard body size. For this reason, GFR estimating equations have been developed in units of ml/min per 1.73 $m^2$. However, because drug clearance is more strongly associated with nonindexed eGFR (ml/min) than indexed eGFR (ml/min per 1.73 $m^2$), in very small or large individuals, this can result in over- or underdosing, respectively, as well as noninitiation of certain medications.[772,773] Nonindexed eGFR can be obtained by multiplying the indexed eGFR results by the person's BSA and dividing by 1.73 $m^2$, or by using an appropriate online calculator.

**Practice Point 4.2.5: Consider and adapt drug dosing in people where GFR, non-GFR determinants of the filtration markers, or volume of distribution are not in a steady state.**

In people with rapidly changing health status, it can be a challenge to estimate the GFR. Serum concentrations of filtration markers may be changing because of changes in true GFR and/or in non-GFR determinants of the marker (Section 1.2). In such settings for people who require medications that are impacted by or could impact GFR, healthcare providers should regularly assess risk, benefits, and value of the medication, and consider higher or lower doses than indicated. Where possible, use medication level testing to guide dosing.[730,774]

### Special considerations

***Dose adjustments in cancer.*** GFR plays a large role in determining anticancer therapy, including anticancer agent selection, dosing, and eligibility for investigational drugs and clinical trials.[775,776] Notwithstanding its lack of validation and its relative inaccuracy compared with other validated eGFR equations, the Cockroft-Gault equation continues to be one of the most commonly used eGFR methods for these people.[775,776] An evaluation of eGFR equation performance against mGFR determined by plasma clearance of $^{51}$Cr-EDTA in 1200 people with solid tumors observed the eGFRcr (CKD-EPI) and the eGFRcr-cys (CKD-EPI) predicted mGFR with greater accuracy than Cockroft-Gault.[136] We advise that the same approach to GFR evaluation described in Section 1.2 be adopted in oncology practice and clinical trials.[136,764] BSA-adjusted eGFR may be indicated for selected specific situations like carboplatin dosing. It is important to consider that non-GFR determinants of both creatinine and cystatin C may be

more profound in people with cancer, and mGFR may be the preferred method to guide the initial dosing for a select group of anticancer drugs including, but not limited to, carboplatin, cisplatin, and methotrexate (Section 1.2).

***Dose adjustment in children/neonates.*** In addition to the usual weight-based dosing for children, specific guidance on drug dosing should be followed for neonates who have lower GFR than those outside the neonatal period.

***Dose adjustment in pregnancy.*** Creatinine decreases physiologically during pregnancy due to glomerular hyperfiltration, and BSA varies. This creates challenges for using GFR or eGFR equations.[54] In such settings for people who require medications that are impacted by or could impact GFR, healthcare providers should regularly assess risk, benefit, and value of medications.

## 4.3 Polypharmacy and drug stewardship

People with CKD are particularly susceptible to polypharmacy due to multiplicity of comorbidities and multiple physician or health system encounters related to those. Most people with CKD not treated with dialysis receive 6–12 different medications per day.[763] Polypharmacy leads to increased pill burden and potential harm due to medication errors and drug-drug interactions. Thus, healthcare providers, including clinical pharmacists, should be diligent in assessing medication appropriateness, number, dose, and potential interactions. Drug stewardship promotes safe medication use throughout the course of therapy. Medications need to be prescribed responsibly, monitored for efficacy and safety, and when no longer required, discontinued.

**Practice Point 4.3.1: Perform thorough medication review periodically and at transitions of care to assess adherence, continued indication, and potential drug interactions because people with CKD often have complex medication regimens and are seen by multiple specialists.**

Medication review is essential for minimizing the occurrence of medication-related problems (e.g., inappropriately high doses and drug interactions) that commonly occur in the CKD population.[777] If a person no longer has an indication for a medication that may contribute to kidney injury (e.g., PPIs), healthcare providers should recognize the opportunity to discontinue the medication. Medication review at each clinical encounter, especially care transitions, is an opportunity to review medication types, interval, and doses especially if the individual has experienced a decline in GFR (e.g., metformin) or physiologic changes that can impact medication volume of distribution (e.g., volume overload and sarcopenia).[778] Figure 46[729] discusses key steps in the medication review process. Three studies have

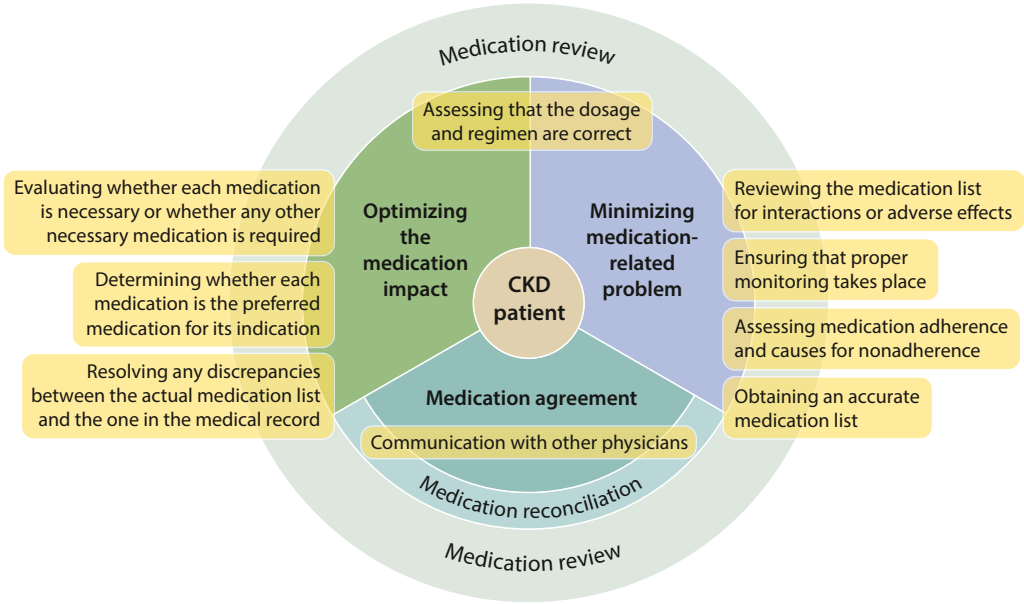


**Figure 46 | Suggested steps in the process of medication review and reconciliation.** Best practices for medication review and reconciliation in people with chronic kidney disease (CKD) include 8 steps[728] and can be summarized as follows: (i) obtain an accurate medication list from the patient; (ii) evaluate whether all medications are medically necessary or whether any other medications is required; (iii) assess whether current therapy represents the "drug of choice" for each indication, individualized for each patient; (iv) evaluate the medication dosage and regimen, taking into consideration related factors such as liver dysfunction, patient size, or weight (e.g., amputation, muscle wasting, and over- or underweight); (v) review the medication list for drug interactions, including drug-drug, drug-disease, drug-laboratory, and drug-food interactions; (vi) ensure that proper monitoring takes place; (vii) determine whether there are any barriers to patient adherence, and evaluate relevant laboratory values; and (viii) identify and resolve any discrepancies between the medications list and the one in the medical record; communication of performed changes in the medication chart with other physicians is necessary given the role of multiple prescribers involved in the care of patients with CKD.[729]

evaluated medication review by clinical practices in people with CKD, observing reductions in the use of inappropriate medications and medication-related problems, both in outpatient and inpatient settings.[765,766] The most frequent reviews involved altering dosage or dose interval and discontinuing NSAIDs. More frequent medication reviews may be needed in older adults with complex medication regimens compared with younger people with CKD.

In the context of good drug stewardship, healthcare providers should be aware of the issue of "prescribing cascade." A prescribing cascade is a sequence of events that begins when an adverse event is misinterpreted as a new medical condition and a subsequent drug is prescribed to treat this adverse event.[779] Before prescribing new medications to address newly reported symptoms, it is important to first assess if the symptoms represent a side effect from an existing medication. An example of a prescribing cascade is as follows: peripheral edema because of calcium channel blocker may be managed by initiation of a new medication (i.e., diuretic), which can lead to additional adverse reactions (e.g., hypokalemia and dizziness).

**Practice Point 4.3.2: If medications are discontinued during an acute illness, communicate a clear plan of when to restart the discontinued medications to the affected person and healthcare providers, and ensure documentation in the medical record.**

Sick day rules have been endorsed as useful guidance to people with CKD in the setting of acute, dehydrating illness. Specifically, patients receive guidance to temporarily stop the following medications: sulfonylureas, ACEi, diuretics/direct renin inhibitors, metformin, ARBs, NSAIDs, and SGLT2i (often described with the acronym SADMANS).[780] However, there is a paucity of evidence to support sick day rules to prevent AKI or other clinically relevant outcomes.[781,782] Instead, data suggest potential harm if people make mistakes in recognizing dehydrating illness or about which drugs to stop and when to restart.[783] Figure 47 shows the steps that must occur correctly for sick day rules to be implemented appropriately. The most reported problem is failure to restart the medication.[784] The plan to restart medications should be detailed in the medical records and clearly communicated to the patients. Patients may additionally benefit from medication review within a month to ensure appropriate medications are restarted.

**Practice Point 4.3.3: Consider planned discontinuation of medications (such as metformin, ACEi, ARBs, and SGLT2i) in the 48–72 hours prior to elective surgery or during the acute management of adverse effects as a precautionary measure to prevent complications. However, note that failure to restart these medications after the event or procedure may lead to unintentional harm (see Practice Point 4.3.2).**

The rationale for temporary discontinuation of certain medications before elective surgery or procedures is to prevent perioperative AKI and other complications such as hypotension or metabolic acidosis or hyperkalemia during the perioperative period.[735] Medications that should be discontinued before elective surgery due to potential perioperative adverse effects are shown in Table 32.[735,736]

There is consistent evidence that withholding RASi is associated with lower risk of perioperative hypotension in various types of surgery and procedures (noncardiac surgery, cardiac surgery, and coronary angiography).[737,785,786] The evidence that withholding RASi would lower perioperative AKI is less consistent as affected by fewer studies with low sample sizes.[787,788] In the surgical context, antihyperglycemic agents such as sulfonylureas, metformin, and SGLT2i would be held because of fasting before the surgery. Case reports, case series, and a systematic review of 47 cases[739,740,789] support the current recommendations that SGLT2i should be withheld at least 3–4 days before the elective surgery.[741,742]

Temporary discontinuation of medications to manage adverse events is indicated in most cases. However, fear for adverse event recurrence often results in failure to resume treatments. In CKD, hyperkalemia or AKI are not uncommon adverse effects of RASi treatment, to which clinical guidelines recommend discontinuation of RASi and therapy reinitiation at low dosages when the event is resolved.[23,29,790,791] Despite this advice, permanent discontinuation of RASi seems to be the most common clinical reaction to occurrence of adverse events.[580,792] Observational studies consistently show that withholding RASi medication compared with continuing treatment after these adverse events is associated with a lower recurrence of adverse events, but conversely a higher risk of MACE and death, for which prevention is one of the main indications for RASi.[502–506] See Section 3.11 on hyperkalemia management.

**Figure 47 | Essential steps for appropriate sick day rule implementation.**

**Table 32 | Medications that should be considered for temporary discontinuation before elective surgeries and potential perioperative adverse events associated with their continued use**

| Medications | Potential perioperative adverse events |
|---|---|
| ACEi/ARB | Hypotension, AKI |
| Diuretics | Volume depletion, AKI |
| SGLT2i | Ketoacidosis (starvation or diabetes) |
| Metformin | Lactic acidosis if AKI occurs |
| Aminoglycosides | Acute tubular necrosis/AKI |
| NSAIDs | AKI, acute interstitial nephritis (AIN) |

ACEi, angiotensin-converting enzyme inhibitor; AKI, acute kidney injury; ARB, angiotensin II receptor blocker; NSAID, nonsteroidal anti-inflammatory drug; SGLT2i, sodium glucose cotransporter-2 inhibitor.

In all these situations, enhanced communication with the patients, and between inpatient and outpatient teams, is necessary to ensure resumption of medications in a timely manner.

#### Special considerations

***Pediatric considerations.*** Many children with CKD with underlying tubular disorders have an obligate urine output irrespective of their hydration status and are at particularly high risk of hypotension and AKI during an acute dehydrating illness. Therefore, temporary discontinuation of medications such as diuretics and RASi that may lead to serious complications of volume depletion, such as hypotension and AKI, should be considered during illnesses. If medications are discontinued during an illness, a clear plan of when to restart the discontinued medications should be communicated to people with CKD and documented in the medical record.

### 4.3.1 Strategies to promote drug stewardship

**Practice Point 4.3.1.1: Educate and inform people with CKD regarding the expected benefits and possible risks of medications so that they can identify and report adverse events that can be managed.**

People with kidney disease have a role in drug stewardship, and given that they may receive medications from non-nephrology healthcare providers, people with CKD should be encouraged to inform those prescribers that they have kidney disease to facilitate consideration of doses and potential side effect of medications. Thus, education and information for people with CKD inclusive for their population (i.e., literacy level and languages) are encouraged. Although brochures and conversations may be useful, interactive electronic health applications have been shown to be acceptable to patients and may lead them to apply the knowledge gained more effectively.[793–797] Practical implementation tips involve printing out the results of the most recent eGFR estimation for the patient to bring along in future healthcare consultations and/or write down a list of ongoing medications to alert other healthcare providers of medication risks and benefits.

A diagnosis of CKD should always be reflected in medical records, as this will alert physicians on the need to consider adjusting or avoiding certain medications or procedures. Under-recognition of CKD diagnoses in medical records is associated with medication errors, including potentially inappropriate prescription of nephrotoxic medications.[729]

**Practice Point 4.3.1.2: Establish collaborative relationships with other healthcare providers and pharmacists and/or use tools to ensure and improve drug stewardship in people with CKD to enhance management of their complex medication regimens.**

Clinical pharmacists are highly qualified experts in medicines and, as part of the multidisciplinary team, can play a pivotal role in improving the quality of care and ensuring patient safety in a range of ways. This includes carrying out structured medication reviews for people with CKD and associated health problems and improving patient safety, outcomes, and value through a person-centered approach.[798–800] In addition and where clinical pharmacists are not available, clinical decision support systems can optimize this process through automation and decision support integrated into the EMRs, supporting drug stewardship through alerts to healthcare providers on the need for dose adjustment to prevent adverse effects. In RCTs enrolling people with CKD, electronic clinical decision support systems have demonstrated efficacy in reducing medication errors, avoiding drug-drug interactions, and improving dose adjustment of medications excreted by the kidneys.[801–806] Recognizing that many of these tools may not be available in all communities, the concepts of regular review and evaluation of medications by a knowledgeable healthcare provider are a critical component of care for people with CKD.

#### Special considerations

***Pediatric considerations.*** Parents and carers should be central to drug stewardship for children with CKD, with increasing involvement from the young person as they move toward transition.

## 4.4 Imaging studies

**Practice Point 4.4.1: Consider the indication for imaging studies in accordance with general population indications. Risks and benefits of imaging studies should be determined on an individual basis in the context of their CKD.**

The use of iodinated radiocontrast media has been associated with the occurrence of AKI, with varying rates reported in observational studies depending on the population studied, the type, route and dose of agent being used, and the definition of nephrotoxicity. The term "contrast-induced AKI" has been traditionally coined to describe this condition,[807] but subsequent research characterizing this entity suggests causal links to be weak,[807–809] and the term "contrast-associated AKI (CA-AKI)" has been suggested instead.

Although there is potential risk for AKI with contrast administration in people with CKD G4–G5, caution should be exercised in withholding contrast treatment or evaluation of a potentially fatal condition solely based on GFR.[810,811] Harm

**Table 33 | Potential risk factors for contrast-associated acute kidney injury**

| Patient-associated | Procedure-associated |
|---|---|
| Reduced GFR, acute or chronic[a] | High-osmolar contrast |
| Diabetes mellitus[b] | Large volume of contrast |
| Reduced intravascular volume | Serial contrast procedures |
| Concomitant nephrotoxic medications | Intra-arterial procedures |

GFR, glomerular filtration rate.
[a]Defined as estimated glomerular filtration rate <45 ml/min per 1.73 $m^2$ with other risk factors or eGFR <30 ml/min per 1.73 $m^2$.
[b]Augments risk in people with underlying kidney function impairment.
Reproduced from Cashion W, Weisbord SD. Radiographic contrast media and the kidney. *Clinical Journal of the American Society of Nephrology*, volume 17, issue 8, pages 1234–1242.[813] https://journals.lww.com/cjasn/fulltext/2022/08000/radiographic_contrast_media_and_the_kidney.20.aspx. Copyright © 2022 by the American Society of Nephrology. The Creative Commons license does not apply to this content. Use of the material in any format is prohibited without written permission from the publisher, Wolters Kluwer Health, Inc. Please contact permissions@lww.com for further information.

may be induced through delaying, not performing, or providing suboptimal diagnostic/therapeutic imaging procedures due to fear of contrast-associated complications to people with reduced GFR.[811,812] Table 33[813] describes the potential causes of CA-AKI identified in available studies that may suggest an approach to people with CKD.

#### 4.4.1 Radiocontrast: intra-arterial and intravenous dye studies

**Practice Point 4.4.1.1: Assess the risk for AKI in people with CKD receiving intra-arterial contrast for cardiac procedures using validated tools.**

The reported risk of CA-AKI is higher with procedures involving arterial administration compared with venous administration of contrast.[814] This difference in risk may be due to differences in patient populations (those who require arterial contrast are likely to have comorbidities that increase the likelihood of AKI) or to differences in the nephrotoxicity of intra-arterial contrast material.

Known risk factors for CA-AKI are advanced age, heart failure, the volume of contrast material, proteinuria, hyperglycemia, and use of RASi.[815] The highest risk for AKI is associated with interventional (rather than diagnostic) coronary angiography (particularly in the setting of acute myocardial infarction). This may relate to the higher volume of contrast used in interventional procedures and hemodynamic instability associated with the clinical situation.[815–817]

**Practice Point 4.4.1.2: The intravenous administration of radiocontrast media can be managed in accordance with consensus statements from the radiology societies in people with AKI or GFR <60 ml/min per 1.73 $m^2$ (CKD G3a–G5) undergoing elective investigation.**

The Work Group refers the reader to the most recent radiology guidelines specifically noting the difference between intravenous and intra-arterial radiocontrast:

- Use of low-osmolality contrast media and iso-osmolarity contrast media
- Use of minimum radiocontrast dose to achieve a diagnostic study
- Withdrawal of nonessential potentially nephrotoxic medications (e.g., NSAIDs, diuretics, aminoglycosides, amphotericin, platins, zoledronate, and methotrexate) in people with AKI or eGFR <30 ml/min per 1.73 $m^2$ for 24–48 hours before and 48 hours after radiocontrast exposure
- In people with eGFR >30 ml/min per 1.73 $m^2$ and without evidence of AKI, metformin need not be stopped before iodinated contrast media (ICM) administration, and there is no need for testing to evaluate GFR afterward. For people with AKI or an eGFR ≤30 ml/min per 1.73 $m^2$, it remains appropriate to stop metformin at the time of or before ICM injection and should not be restarted for at least 48 hours and only then if GFR remains stable and the ongoing use of metformin has been reassessed by the clinical team.[818]
- Given the lack of strong evidence demonstrating that continuing RAASi is beneficial, referring healthcare providers should consider withholding RAASi in people at risk for ≥48 hours before elective contrast-enhanced CT to avoid the potential for hypotension and hyperkalemia should CA-AKI develop. RAASi may be restarted if CA-AKI does not occur or after the return of GFR to baseline.
- Consideration of avoiding dehydration for people not undergoing dialysis and who have eGFR <30 ml/min per 1.73 $m^2$ or AKI and receiving intravenous contrast.[815,819]
- Use of N-acetylcysteine, ascorbic acid, furosemide, dopamine, fenoldopam, or calcium channel blockers as preventative measures of CA-AKI has not been shown to be a consistent benefit.[813]
- Prophylactic pericontrast hemodialysis has been shown to be potentially harmful and is not recommended.[813]

**Special considerations**

***Global access to contrast agents.*** The preference of contrast agent may depend on availability and cost, particularly in lower-income countries and lower-middle–income countries.

#### 4.4.2 Gadolinium-containing contrast media

Gadolinium chelates used during magnetic resonance imaging has previously been reported to cause nephrogenic systemic fibrosis (NSF) before 2010, and the mechanisms have been articulated.[820] Note that incidence of this condition has not been reported later than 2012, thus raising the question as to the true risk of this condition.[821]

**Practice Point 4.4.2.1: For people with GFR <30 ml/min per 1.73 $m^2$ (CKD G4–G5) who require gadolinium-containing contrast media, preferentially offer them American College of Radiology group II and III gadolinium-based contrast agents.**

People who are at greatest risk for NSF include those with AKI, undergoing KRT, and those with CKD G4–G5. Most unconfounded cases have been associated with American College of Radiology group I gadolinium-based contrast

media (e.g., gadodiamide, gadopentetate dimeglumine, and gadoversetamide), and there is additional risk with repeated doses.[822,823]

Hence, in people with GFR <30 ml/min per 1.73 $m^2$, the use of newer linear and macrocyclic gadolinium-based contrast media such as gadobenate dimeglumine, gadobutrol, gadoteridol, gadoterate meglumine, and gadoxetate disodium should be preferred.[824,825]

### Special considerations

***Global access to gadolinium-contrast agents.*** There are cost implications in lower-income countries and lower-middle–income countries as the nonlinear-chelated preparations are more expensive.

***Pediatric considerations.*** Considerations specific to the use of gadolinium preparations in young children and neonates must also be contemplated in addition to the general admonishments against their use in situations of GFR <30 ml/min per 1.73 $m^2$. In particular, the FDA currently does not license any gadolinium-based contrast media product for use in children <2 years of age and, likewise, the European Medicines Agency cautions against the use of any gadolinium-based contrast agents in a child <1 year of age. The risk of NSF in pediatric populations appears to be low, but data are limited.[822,823]

In recognition of the inability to accurately measure GFR in the neonate and, by extension, the clearance of compounds such as gadolinium, all nephrologists and radiologists must exercise caution in terms of use of gadolinium-based contrast media in this potentially high-risk population, and all other imaging modalities should be considered before choosing one requiring gadolinium exposure. Although not based on specific evidence, some have suggested the avoidance of high-risk gadolinium agents in very young children (e.g., neonates younger than 4 weeks of age).[826]

Moreover because of kidney immaturity in fetuses, neonates, and infants, this population (and consequently pregnant women because of the risk to the fetus) is considered potentially at risk for NSF.[827] However, although the data are limited, the number of reported cases of NSF in the pediatric population is lower than in the adult population.[828] There is no convincing evidence that pediatric populations have an increased risk compared with adults. The risk of NSF in pediatric patients appears to be low, but data are limited.[823]

**For research recommendations, please see Chapter 6: Research recommendations.**